The Quantum Ground:

Sequenced Excitations, Information Emergence, and Cross-Scale Coherence

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Abstract

The Imprint Hypothesis established a field-first framework in which quantum fields constitute the persistent substrate of reality, particles emerge as localized excitations, and wave-like phenomena arise as contextual imprints generated through particle-field interactions. Subsequent work on Detector-Plane Imaging (DPI) and the Quantum Measurement Stack (QMS) extended this framework by introducing measurement-plane coherence diagnostics and detector-plane causality as operational foundations for quantum measurement.

This paper introduces the concept of the **Quantum Ground**, an observational ontology that unifies spacetime, quantum fields, and energy as inseparable aspects of a common physical substrate. Within this framework, particles are interpreted as localized excitation events, information emerges through the sequencing of excitation histories, and macroscopic structure arises through the preservation of sequenced imprints across scales.

No new physical laws are proposed. The framework remains fully compatible with Quantum Field Theory (QFT), decoherence theory, the Imprint Hypothesis, DPI, and QMS. Its contribution is the introduction of a unified vocabulary linking quantum phenomena, information emergence, measurement, and macroscopic reality through the principle of Cross-Scale Coherence.

Keywords: Quantum Foundations, Quantum Field Theory, Imprint Hypothesis, Quantum Ground, Detector-Plane Imaging, Quantum Measurement Stack, Information Emergence, Cross-Scale Coherence

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1. Introduction

Modern physics successfully describes nature across a vast range of scales. Quantum Field Theory explains elementary particles as excitations of underlying fields, while statistical mechanics, chemistry, biology, and cosmology describe increasingly complex emergent structures.

Despite their success, these descriptions often appear conceptually fragmented.

The present work explores whether a single observational ontology can provide continuity across these scales without modifying established physics.

Building upon the Imprint Hypothesis, Detector-Plane Imaging (DPI), and the Quantum Measurement Stack (QMS), we introduce the concept of the Quantum Ground and propose that information emerges through the sequencing of excitation events occurring within this common substrate.

The objective is not to replace existing mathematical frameworks but to provide a coherent observational language that remains consistent from quantum excitations to macroscopic reality.

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2. The Quantum Ground

We begin with the proposition that physical reality is fundamentally characterized by the inseparable coexistence of:

$$Q_G = (\text{Spacetime}, \text{Quantum Fields}, \text{Energy})$$

where:

- Spacetime provides geometry and causal structure.
- Quantum fields provide identity and interaction channels.
- Energy provides excitation potential.

The Quantum Ground is not introduced as a new physical field or hidden variable.

Rather, it represents a unifying description of structures already present in modern physics.

Within this framework, the vacuum is not interpreted as empty space but as the lowest accessible excitation state of the Quantum Ground.

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3. Localized Excitation Events

Particles are treated as localized excitation events within the Quantum Ground.

A quantum particle is defined as:

> A localized, identity-bearing excitation of a quantum field occurring within spacetime.

Examples include:

- Electron-field excitation → electron
- Electromagnetic-field excitation → photon
- Quark-field excitation → quark

Particle identity is inherited directly from the excited field.

This preserves particle realism while remaining fully compatible with standard QFT.

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4. Contextual Imprints

The Imprint Hypothesis proposes that wave-like phenomena do not constitute a separate ontology.

Instead, interference, diffraction, tunneling, and related phenomena emerge as contextual imprints generated through particle-field interactions.

The process may be represented conceptually as:

Localized Excitation Event

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Interaction Dynamics

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Contextual Imprint

Wave behavior therefore reflects interaction history rather than an independent physical entity.

Measurement fixes observable imprints but does not create particles.

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5. Sequenced Excitations and Information

The central proposal of this work is that information emerges through the sequencing of excitation events.

Energy alone does not determine structure.

Two systems may contain similar energy content while exhibiting dramatically different organizational complexity due to differences in excitation history.

We therefore propose:

> Information is preserved sequencing.

Information emerges from ordered relationships between excitation events occurring within the Quantum Ground.

This principle applies across scales:

- Quantum coherence
- Atomic organization
- Molecular chemistry
- Biological systems
- Macroscopic structures

Complexity emerges not from additional physical ingredients but from the preservation and accumulation of excitation sequences through time.

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6. Cross-Scale Coherence

We introduce the principle of Cross-Scale Coherence:

> A valid ontology should remain self-consistent when describing the same physical system across all observable scales.

Consider a table.

The same object may be described as:

Table

↓

Material Structure

↓

Molecular Organization

↓

Atomic Configuration

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Quantum Particles

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Field Excitations

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Quantum Ground

These descriptions are not competing realities.

They are different resolutions of the same underlying system.

Cross-Scale Coherence requires conceptual continuity between all descriptive levels.

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7. Measurement and Detector-Plane Causality

The Quantum Measurement Stack identifies the detector plane as the effective measurement boundary.

Within the QMS framework:

Quantum Ground

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Localized Excitation

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Contextual Imprint

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Detector-Plane Registration

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Measurement Record

Information discarded at the detector plane cannot be recovered through later processing.

This localizes irreversibility within physical measurement systems rather than within observers or consciousness.

Human observation occurs only after detector-plane registration.

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8. Position Within the Existing Research Program

The Quantum Ground framework should not be viewed as a replacement for previous work.

Rather, it extends and unifies concepts developed through:

Imprint Hypothesis

- Field primacy
- Particle realism
- Wave contextuality

Detector-Plane Imaging (DPI)

- Measurement-plane coherence evolution
- Pre-discretization coherence diagnostics

Quantum Measurement Stack (QMS)

- Detector-plane causality
- Localization of irreversible information loss
- Measurement integrity benchmarking

The present work introduces:

- Sequenced Excitations
- Information as Preserved Sequencing
- Cross-Scale Coherence
- Quantum Ground Ontology

as an observational framework connecting these concepts across scales.

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9. Discussion

The primary contribution of this work is conceptual continuity.

The framework suggests that:

- Particles are localized excitation events.
- Wave-like behavior emerges as contextual imprints.
- Information emerges through sequencing.
- Measurement fixes surviving imprints.
- Macroscopic reality emerges through cross-scale coherence.

No new dynamical laws are introduced.

The framework serves as an organizational and observational ontology intended to provide continuity between quantum processes and emergent structure.

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10. Conclusion

The Quantum Ground framework unifies spacetime, quantum fields, and energy within a single observational ontology.

Within this framework:

- Quantum particles are localized excitation events.
- Wave phenomena are contextual imprints.
- Information emerges through sequenced excitations.
- Measurement fixes surviving imprints.
- Macroscopic reality emerges through cross-scale coherence.

The framework remains compatible with Quantum Field Theory, the Imprint Hypothesis, Detector-Plane Imaging, and the Quantum Measurement Stack while providing a unified vocabulary spanning quantum, microscopic, macroscopic, and cosmological domains.

Project Resources

Publications

Imprint Hypothesis: https://zenodo.org/records/20470449

Detector-Plane Imaging: https://zenodo.org/records/18994419

Quantum Measurement Stack: https://zenodo.org/records/19153014

QMCTB Benchmark Artifacts: https://zenodo.org/records/19147730

Source Code and Repositories

Project Repository: https://github.com/srikarr20

References

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